

Step 3: Sequence Generation

Continuous 3D Morphology Synthesis

The Objective

Once the Backward Diffusion model predicts that two fragments connect, the **Sequence Generator** predicts how.

It generates a continuous sequence of 3D coordinates $X_{\{1:L\}}$ that bridges the topological gap, conditional on the invariant contexts of the two fragments.

The Denoising Process (DDPM)

The model functions as a continuous Denoising Diffusion Probabilistic Model. - **Input:** Pure Gaussian noise $N(0, I)$ sampled into a 3D tensor of shape (Batch, MaxLength, 3). - **Conditioning:** The invariant Context Vectors (c_1, c_2) extracted from the Backward Diffusion Model. - **Output:** A smooth, biologically plausible 3D curve tracing a neurite path.

Layer-wise Architecture: 1D-Chain EGNN

1. **Initialization:** $h_0 = \text{TimeMLP}(t) + \text{ContextProj}(c_1 \oplus c_2) + \text{SinusoidalPos}(0..L)$
 2. **1D-Chain Path Encoding:** 4 unrolled **EGNN Layers**. The nodes are connected as a 1D chain ($i \rightarrow i+1$).
 3. **Equivariant Noise Prediction:** The EGNN directly predicts the rotation-equivariant coordinate noise update: $\Delta x = x_{\text{out}} - x_t$.
 4. **Node Type Head:** Predicts invariant categorical intents: Continue (0), Branch (1), Terminate (2).
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Optimization: Morphological Losses

The generation is optimized using two primary losses: - **Coordinate Loss (Weight: 10.0):** Masked Mean Squared Error (`nn.MSELoss`) over the predicted coordinate noise vs the added Gaussian noise. The heavy weight prioritizes smooth, accurate paths. - **Node Type Loss (Weight: 0.5):** Masked Cross-Entropy Loss for structural intents.

Padding masks ensure that variable-length branches do not corrupt the loss gradient.